

THE EFFECTS OF SWISS BALL EXERCISE ON PAIN AND FUNCTION IN PATIENTS WITH LOW BACK PAIN: A SYSTEMATIC REVIEW



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BACKGROUND

- Low Back Pain (LBP) is a leading cause of disability worldwide and imposes a significant burden on individuals and healthcare systems.
- Exercise is a key component in the management of LBP.
- Swiss ball exercise (SBE) is increasingly used in physiotherapy to improve core stability, muscle strength, balance and posture.
- This review aims to evaluate the effects of Swiss ball exercise on pain and function in patients with LBP.



METHODS

Design : Systematic Review
Database : CINAHL, MEDLINE, AMED, APA PsycINFO
Search period : Until September 2024
Inclusion criteria:

- Randomized Controlled Trials (RCTs)
- Adults (≥18 years) with non-specific LBP
- Swiss ball exercise as intervention
- Outcomes: Pain, and/or Function
- Published in English

Exclusion criteria:

- Studies with other main interventions
- Non-RCT studies
- Inadequate outcome data

Quality assessment : PEDro Scale
Data extraction : One independent reviewer



WHAT IS SWISS BALL EXERCISE?

Swiss ball exercise involves performing movements using an unstable ball to activate core muscles, improve spinal stability, flexibility, strength and neuromuscular control.



Pelvic Tilt



Bridge



Bird Dog



Side Plank



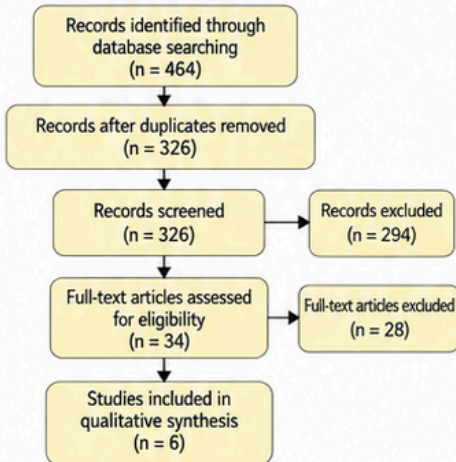
RESEARCH QUESTION

What are the effects of Swiss ball exercise on pain and function in patients with low back pain?



RESULTS

SEARCH YIELD



OBJECTIVE

To systematically review the effectiveness of Swiss ball exercise in reducing pain and improving function in patients with low back pain.

RESULT SUMMARY (n = 6 RCTs)

PAIN



All 6 studies reported significant reductions in pain intensity (measured by VAS/NPRS) compared to control groups.

FUNCTION



All studies demonstrated significant improvements in functional outcomes (e.g., Oswestry Disability Index, Roland-Morris Disability Questionnaire).

OVERALL FINDING



Swiss ball exercise shows beneficial effects in reducing pain and improving function in patients with low back pain.



IMPLICATION FOR PRACTICE

- Swiss ball exercise can be safely incorporated in physiotherapy programs for LBP.
- Offers a cost-effective and low-risk option to improve core stability, reduce pain and enhance daily function.
- Can be used in various clinical settings and adapted to individual needs.



STRENGTHS & LIMITATIONS

STRENGTHS:

- ✓ Included only RCTs
- ✓ Comprehensive search strategy
- ✓ Quality assessment using PEDro scale

LIMITATIONS:

- ✗ Small number of studies
- ✗ Heterogeneity in intervention protocols
- ✗ Short-term follow-up in most studies



CONCLUSION

Swiss ball exercise is effective in reducing pain and improving function in patients with low back pain. It should be considered as part of a structured physiotherapy intervention.



TAKE-HOME MESSAGE

Incorporate Swiss ball exercise into your LBP rehabilitation program to reduce pain, improve function and promote better quality of life. Consistency, Progression and proper technique are the keys to success!



REDUCE PAIN



IMPROVE FUNCTION



ENHANCE QUALITY OF LIFE

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- (And other references as per included studies).